

CYCLISTIC BIKE-SHARE

Data Analysis Report

How Members and Casual Riders Differ

May 2025 – April 2026 | 5,535,455 Rides Analyzed

5.5M Total Rides	12 Months of Data	64% Annual Members
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Tools Used: BigQuery (SQL) · Tableau Public

1. Executive Summary

This report analyzes 12 months of Cyclistic bike-share data (May 2025 – April 2026) to answer the business question: How do annual members and casual riders use Cyclistic bikes differently? The findings provide a data-driven foundation for a targeted marketing strategy to convert casual riders into annual members.

Key Findings at a Glance

Finding	Casual Riders	Annual Members	Implication
Ride Duration	19.2 min avg	11.8 min avg	Casuals = leisure; Members = commute
Weekly Peak	Saturday (398K)	Thursday (589K)	Casuals = weekends; Members = weekdays
Seasonal Drop	93% (Jan vs Aug)	75% (Jan vs Aug)	Casuals are weather-dependent
Top Stations	Navy Pier, Millennium Park	Spread city-wide	Casuals prefer tourist spots
Bike Preference	66% electric	65% electric	Classic bike rides: casuals 2.1x longer

2. Data Overview & Methodology

Data Source

Data: Divvy Bike-Share public dataset (Motive International Inc.), licensed for public use.

Period: May 2025 – April 2026 (12 consecutive months)

Scope: City of Chicago, Illinois, USA

Data Cleaning Summary

All 12 monthly CSV files were uploaded to Google BigQuery and cleaned using SQL. The following issues were identified and resolved:

- Duplicate ride IDs: 0 found across all 12 months
- Missing GPS coordinates (end_lat/end_lng): removed rows where end location was null
- Rides under 60 seconds: removed as test rides or accidental unlocks
- Rides over 24 hours (1,440 minutes): removed as likely unreturned bikes
- Timestamp format inconsistencies in one file: corrected using PARSE_TIMESTAMP()

Metric	Value
Raw rows (all 12 months)	~5,700,000
Rows removed (cleaning)	~164,545
Final clean rows	5,535,455
Cleaning rate	~2.9%

New Columns Created

- ride_length_minutes: TIMESTAMP_DIFF(ended_at, started_at, MINUTE)
- day_of_week: EXTRACT(DAYOFWEEK FROM started_at) — 1=Sunday through 7=Saturday

3. Analysis Findings

Finding 1: Ride Duration

Casual riders take significantly longer trips than annual members, suggesting fundamentally different use cases.

19.2 min Casual Average Ride	11.8 min Member Average Ride	63% Longer (Casual vs Member)
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When broken down by bike type, the leisure nature of casual rides becomes even clearer:

Bike Type	Casual Avg Duration	Member Avg Duration	Casual/Member Ratio
Electric Bike	14.3 min	10.9 min	1.31x
Classic Bike	28.7 min	13.4 min	2.14x

Casual riders on classic bikes average 28.7 minutes, nearly half an hour per ride, strongly indicating leisure and exploratory behaviour rather than point-to-point commuting.

Finding 2: Weekly Ride Patterns

The weekly distribution of rides reveals a clear behavioural split between the two groups.

Day	Casual Rides	Member Rides	Casual % of Day Total
Sunday	325,077	383,108	46%
Monday	223,743	503,442	31%
Tuesday	220,387	569,292	28%
Wednesday	220,429	567,025	28%
Thursday	258,310	589,185	30%
Friday	306,955	527,383	37%
Saturday	398,006	443,113	47%

- Casual riders peak on Saturday (398,006 rides) and Sunday (325,077) — 37% of casual rides fall on weekends
- Members peak on Thursday (589,185) with consistently high weekday usage — 77% of member rides are Mon-Fri
- On weekdays, members outnumber casuals by 2:1 or more

Finding 3: Seasonal Trends

Both groups show seasonal patterns, but casual riders are far more sensitive to weather and time of year.

Month	Casual Rides	Member Rides	Casual Share
January	23,876	109,855	18%
February	40,072	157,230	20%
March	84,796	223,643	27%
April	127,025	308,470	29%
May	175,648	313,994	36%
June	278,675	379,517	42%
July	308,429	430,392	42%
August	323,523	443,125	42%
September	254,714	440,951	37%
October	214,373	414,088	34%
November	94,689	251,912	27%
December	27,087	109,371	20%

- Casual rides drop 93% from August peak (323,523) to January trough (23,876)
- Member rides drop only 75% from August (443,125) to January (109,855)
- The May–September window accounts for 72% of all casual rides

This seasonality tells us: summer is the prime conversion window. Casual riders are most engaged and most reachable between May and September.

Finding 4: Station Hotspots

The geographic patterns clearly separate tourist/leisure use (casual) from utility use (member).

Rank	Casual Top Stations	Rides	Location Type
1	Navy Pier	32,173	Tourist attraction
2	DuSable Lake Shore Dr & Monroe St	31,083	Lakefront park
3	Michigan Ave & Oak St	22,257	Shopping / tourism
4	DuSable Lake Shore Dr & North Blvd	19,273	Lakefront park

Rank	Casual Top Stations	Rides	Location Type
5	Streeter Dr & Grand Ave	18,910	Lakefront / tourism
6	Millennium Park	18,566	Tourist attraction
7	Shedd Aquarium	16,556	Tourist attraction

In contrast, member stations are distributed throughout the city along commuter corridors and transit hubs, not concentrated at tourist attractions. This confirms that members use bikes for daily transport.

Finding 5: Bike Type Preferences

Both groups show similar overall bike type preferences, but the duration patterns reveal important differences.

Bike Type	Casual Rides	Casual %	Member Rides	Member %
Electric Bike	1,294,903	66%	2,310,294	65%
Classic Bike	658,004	34%	1,272,254	35%

4. Marketing Recommendations

Based on the data analysis, three targeted recommendations are proposed to convert casual riders into annual members.

Recommendation 1: Weekend & Summer Membership Campaign

Data Supporting This Recommendation

- Casual peak: Saturday 398,006 rides (highest day)
- Casual seasonal peak: August 323,523 rides
- Summer (May-Sep) = 72% of all casual rides

Proposed Action

Launch a 'Weekend Rider to Annual Member' upgrade campaign running from May through August. Target casual riders who have used the service 3 or more times within a rolling 30-day window with an in-app notification offering a time-limited discounted annual membership.

Display promotional messaging at the top 10 casual stations during weekend hours (Friday evening through Sunday). The campaign should emphasise value during the season casual riders are already most engaged.

Expected Impact

Converting even 5% of the 1.95M casual riders to annual members would add approximately 97,500 new annual members a significant revenue uplift.

Recommendation 2: Digital Ads at Casual Hotspot Stations

Data Supporting This Recommendation

- Top 3 casual stations alone account for 81,313 rides
- All top casual stations are tourist/leisure spots on or near the lakefront
- These stations have very low member usage proportionally

Proposed Action

Install digital display boards and QR-code posters at the top 10 casual stations (Navy Pier, Millennium Park, Shedd Aquarium, DuSable Lakefront, etc.). Displays should show real-time cost comparisons:

- 'Today you spent \$X. Annual membership pays for itself in just 8 rides.'
- 'Unlimited weekend rides for \$Y/month. No top-up needed.'

QR codes should link directly to a frictionless membership sign-up page, optimized for mobile.

Expected Impact

Capturing riders at the moment of use when they are already engaged and paying is the highest-conversion marketing opportunity available.

Recommendation 3: Reframe Membership as a Leisure Lifestyle

Data Supporting This Recommendation

- Casuals average 19.2 min per ride — not a quick commute trip
- Classic bike rides by casuals average 28.7 min — exploratory, enjoyable rides
- Current top casual stations are leisure destinations, not offices or transit hubs

Proposed Action

Audit all existing Cyclistic marketing materials. Current messaging likely focuses on commuting efficiency and daily transport (appealing to members). A parallel campaign should reframe annual membership as a leisure lifestyle choice:

- Campaign theme: 'Ride longer, explore more, pay less'
- Imagery: lakefront sunsets, weekend exploration, couples and friends riding to office buildings or transit stations
- Channels: Instagram and TikTok (weekend leisure audiences) in addition to commuter channels

This does not replace the commuter message, it adds a leisure membership pitch aimed specifically at the casual rider segment that data shows is already riding for fun.

5. Conclusion

The data tells a consistent and actionable story. Casual riders and annual members are not using Cyclistic bikes in the same way they are, in many respects, using a fundamentally different service that happens to run on the same infrastructure.

Casual riders ride longer, on weekends, during summer, from tourist stations, for leisure. Members ride shorter, on weekdays, year-round, from commuter hubs, for efficiency.

This distinction is the key to conversion. The marketing strategy should not try to convince casual riders to change their behaviour, it should show them that their existing leisure behaviour is worth paying for consistently, because membership is better value than paying per ride.

Summary of Recommendations

	Recommendation	Target Window	Priority
1	Weekend & Summer Membership Campaign	May – August	HIGH
2	Digital Ads at Casual Hotspot Stations	Year-round	HIGH
3	Reframe Membership as Leisure Lifestyle	Year-round	MEDIUM

— End of Report —

Cyclistic Bike-Share Case Study